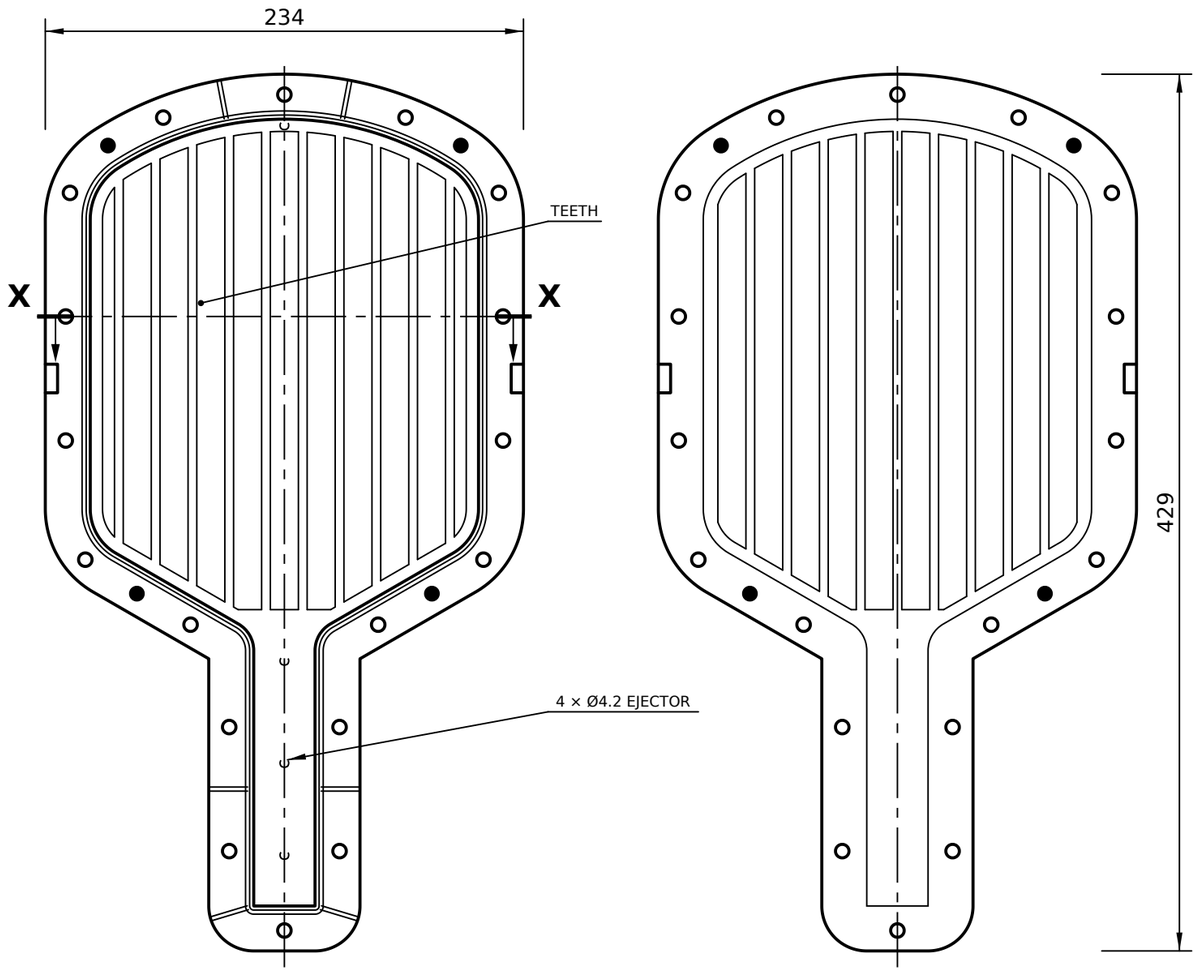
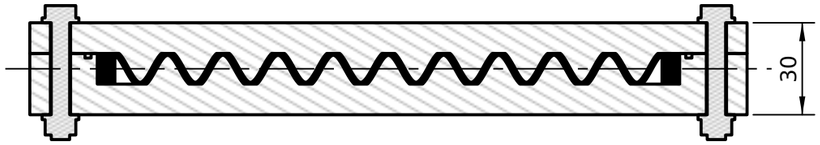


TRAY (BOTTOM HALF)

SCALE 1:3 · TEETH = BOTTOM FLUTES

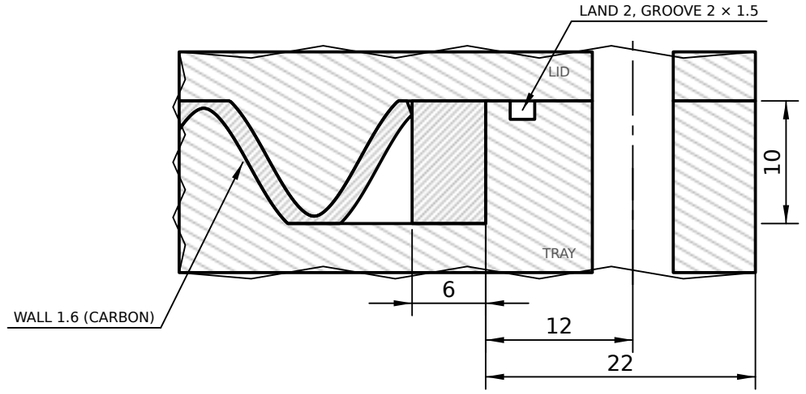
LID, UNDERSIDE

SCALE 1:3 · TEETH = TOP FLUTES



SECTION X–X

SCALE 1:2 · CLOSED MOULD, MOULDED RIB SUPPORT BLACK, M6 BOLT



DETAIL Y

SCALE 2:1 · TEETH INTERLOCK; THE GAPS ARE THE WALLS AND THE FRAME

NOTES

1. TRAY (1) + LID (2): output/mould/hybrid_mould_bottom / _top (.stl / .step). THE CAVITY BETWEEN THEM IS THE RIB SUPPORT (SHEET 3); THE TEETH ARE ITS FLUTES. THE LID SEATS ON THE TRAY RIM: CORE HEIGHT 10, WALLS 1.6.
2. PRINT BOTH BACK DOWN, TEETH UP: ≥ 5 WALLS, ≥ 40 % INFILL, 0.12–0.16 LAYERS FOR THE TEETH, IRONING ON FLAT TOPS. NEEDS A ≥ 440 mm BED, OR MACHINE THE STEP FROM ALUMINIUM. ABOUT 1.3 + 1.0 kg OF PLA AT 100 %.
3. PLA SOFTENS NEAR 55 °C: ROOM-TEMPERATURE-CURE EPOXY ONLY. FOR A WARM CURE PRINT IN PETG, ASA OR PA-CF.
4. PREPARE: SAND THE CAVITY FACES (P400 → P800), SEAL (EPOXY COAT), WAX ×3 + PVA. PRESS THE 4 DOWEL PINS (4) INTO THE TRAY. PLUG THE EJECTOR HOLES FROM BELOW WITH TAPE.
5. FILL (FORGED CARBON): CHOP CARBON TOW TO 10–25 mm, WET IT WITH EPOXY (≈ 40 % FIBRE BY VOLUME). PACK THE WALL GAPS AND THE FRAME FIRST, THEN THE SOLID HANDLE ZONE; OVERFILL BY ≈ 10 %.
6. CLOSE: LID OVER THE DOWELS, BOLTS (3) FINGER-TIGHT, THEN PRESS BETWEEN 2 FLAT, STIFF PLATENS OR IN A PRESS UNTIL THE LID SEATS ON THE RIM; TIGHTEN THE BOLTS. EXCESS RESIN LEAVES BY THE OVERFLOW GROOVE AND THE VENTS.
7. CURE (24 h AT ROOM TEMPERATURE), SPLIT AT THE PRY NOTCHES, LIFT THE LID STRAIGHT UP, PUSH THE PART OUT WITH THE 4 EJECTOR PINS. TRIM THE FLASH, DRILL AND FIT THE M3 INSERT (SHEET 3). THE PART WEIGHS ≈ 257 g.
8. TRIAL FIRST: PRINT THE RIB SUPPORT STL IN PLA AND CLOSE THE EMPTY MOULD ON IT – IT MUST SEAT FULLY.

MOULD HARDWARE

ITEM	PART	QTY
1	Tray, hybrid_mould_bottom.stl	1
2	Lid, hybrid_mould_top.stl	1
3	Hex bolt M6 × 50, 2 washers, nut	18
4	Dowel pin Ø6 × 16, steel (ISO 2338)	4
5	Ejector pin Ø4 × 40 (or a drill shank)	4
6	Release wax + PVA; flat platens + clamps or a press	a/r

RIB SUPPORT MOULD – HYBRID

RIB SUPPORT MOULD · TRAY + LID, TEETH INTERLOCK · 234 × 429 mm

DRAWN	DATE	CHECKED	APPROVED
Claude Code	2026-10-25		

DRAWING No.	REV	SHEET	SIZE	SCALE	UNITS
PB-HY-001	E	4 / 4	A3	1:3, 1:2, 2:1	mm

PROJECTION	MATERIAL	STATUS
	PLA / PETG (NOTE 2)	ISSUED FOR REVIEW